# Reliable procurement through n8n.

An enterprise integration platform with human approval, durable financial state, and recoverable service orchestration.

| 01 | 02 | 03 | 04 |
|---|---|---|---|
| React | NestJS | RabbitMQ + BullMQ | n8n |
| Employee / manager / admin | Authoritative API + roles | Durable dispatch / retry | Versioned HTTP workflows |

EXECUTION PATH

## A short-lived workflow around a durable decision.

### Collection / procurement-collect-v1

An event triggers finance and supplier HTTP calls. The authenticated callback validates quotes, atomically reserves the lowest eligible amount, and assigns an approver.

### Finalisation / procurement-finalize-v1

Approval is saved in PostgreSQL between webhooks. A second event starts n8n; NestJS creates one purchase order and converts reserved funds into spent funds.

**DECISION BOUNDARY** Manager approval is persisted, not held as a waiting n8n job.

CORE ENGINEERING

## Responsibility stays with the right service.

### PostgreSQL / Prisma

Source of truth for requests, department budgets, quotes, approvals, orders, audit entries, workflow versions and execution attempts. Serializable transactions and unique constraints prevent double spend.

### RabbitMQ / Redis / BullMQ

Transactional outbox and broker confirms protect publishing. Inbox receipts prevent duplicate dispatch. BullMQ performs five webhook attempts with exponential backoff and reconstructs deadline jobs from persisted timestamps.

### NestJS / shared contracts

DTO validation, RBAC, Swagger / OpenAPI and callback auth govern business mutations. REST contracts are shared with the React frontend and integration schemas.

### Observability / deployment

Pino and OpenTelemetry feed correlation-aware traces to an OTLP collector and Jaeger. Docker Compose runs the local stack; optional OpenTofu provisions Docker resources only.

# Failure is part of the workflow.

A request must stay correct when a webhook repeats, finance rejects, approval expires or a supplier becomes unavailable.

CONSISTENCY MODEL

## Idempotency, compensation and recovery.

### Outbox & broker confirms

The event and procurement record commit together. Publisher redelivery is safe after a crash between confirm and transaction commit.

### Attempt-scoped callbacks

Callback keys and unique execution/attempt/phase receipts reject duplicates, stale attempts and wrong-phase results.

### Financial compensation

Rejection and expiry release reservations. Retrying failed approved finalisation atomically re-reserves the approved quote before ordering.

### Execution deadlines

PostgreSQL deadlines are reconciled into delayed BullMQ jobs; missing callbacks become visible timed-out exceptions.

CONTROLS

## Human roles, bounded credentials and evidence.

### Access and integration security

Employee ownership checks, assigned-manager approval, admin-only recovery/configuration. Separate internal webhook token and phase callback key. AES-256-GCM secret storage; local API key or simulated OAuth2 client-credentials exchange.

### Versioning and history

Workflow rules are published as immutable versions. Each request retains its starting definition, correlation ID, attempts and prior audit results. Database triggers block ordinary updates of protected history.

VERIFIED RESULTS

## Executed local evidence / 3 October 2026.

| | |
|---|---|
| Completed | Local OAuth2 exchange, collected quotes, manager approval, single order, budget moved to spend. |
| Rejected | Manager rejection, no purchase order, released reservation. |
| Insufficient budget | No approval or order, and no unauthorised reservation. |
| Recovered | Simulated supplier failure, admin retry, new attempt, then manager-authorised completion. |

**VERIFICATION SCOPE**

Nine services started; generated OpenAPI exposed 21 routes; smoke and invariant suites passed. Real n8n webhooks and RabbitMQ receipts were observed. No external supplier or production cloud deployment was tested.